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RESEARCH ARTICLE

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# The impact of regulatory workplace safety inspections on workers' compensation claim rates

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## Abstract

**Background:** Studies on the impact of workplace safety inspections on work injuries have found mixed effectiveness. Most studies are from the United States, examining Occupational Health and Safety Administration (OSHA) regulatory inspections in manufacturing firms with more than 10 employees. This study examines whether regulatory inspections in Alberta, Canada, result in reductions in workers' compensation claims rates for inspected firms relative to comparable non-inspected firms.

**Methods:** Firm and claim-level data from the Workers' Compensation Board of Alberta were linked with regulatory enforcement data from the Government of Alberta for construction, manufacturing, and transportation firms with at least one full-time employee for 37 consecutive months. A matched difference-in-differences study design was used to estimate changes in lost-time claim rates for work-related injuries and musculoskeletal diseases of inspected and comparable non-inspected firms between the year pre-inspection and 2 years, post-inspection, controlling for firm-level characteristics.

**Results:** Inspections were not effective in reducing firm-level claim rates, with the exception of transportation firms with more than one inspection experiencing a 28% decrease in their claim rate in the second year post-inspection, relative to the change in non-inspected firms. In construction, inspected firms experienced a 12% increase in their claim rate in the first year post-inspection. No effect was observed in the manufacturing sector.

**Conclusions:** Regulatory workplace safety inspections in Alberta generally do not result in greater reductions in firm-level claim rates in the construction, manufacturing, and transportation sectors. Inspections alone may not be sufficient to induce compliance or hazard management changes that lead to reductions in firm-level injuries.

KEYWORDS

enforcement, injuries, regulatory effectiveness, workers' compensation claims

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## 1 | INTRODUCTION

Regulatory enforcement activities, such as workplace consultations, inspections, citations and penalties, are key tools for occupational health and safety (OHS) agencies to ensure compliance with OHS regulations and reduce workplace injuries and fatalities. Research from systematic reviews has found that inspections overall, and with citations and/or penalties, are associated with increased compliance and reduced firm-level injury rates.[1,2] However, this research has largely been conducted in the United States or other countries where the OHS regulatory environment is different than Canada, and findings may not be generalizable to Canadian jurisdictions.

In Canada, OHS legislation outlines the general rights and responsibilities of the employer, the supervisor and the worker through an Act or statute and related regulations. Each of the 10 provinces, three territories and the federal government has its own OHS legislation.[3] As a result, there are variations in how OHS legislation and regulations are formulated and enforced across jurisdictions. During inspections, inspectors may find violations of the regulation, in which case they may issue orders (typically called citations in the United States), written directives for a person or firm to take measures to comply with legislation or regulation or levy financial penalties against employers for safety infractions. In the United States, the jurisdiction of OHS is shared between the Federal Occupational Health and Safety Administration (OHSA) and various state OHSA plans, which include workplace safety inspections as part of their prevention and enforcement programs. The main differences between the regulatory environment in Canada and the United States are that programmed inspections by OHSA are limited to firms with more than 10 employees, that rates of inspection are generally lower in the United States than in Canada, and that financial penalties when inspections are more common.[4]

Only three studies have examined the effectiveness of regulatory inspections on work-related injuries in Canadian jurisdictions. An Ontario study examined the effect of inspections and consultations on firm-level injury rates after randomizing manufacturing firms into intervention and control groups.[5] A British Columbia (BC) study examined the effect of inspections and inspections with orders on firm-level injury rates using quasi-experimental design methods.[4] Both studies found that inspections generally did not lead to reductions in firm-level injury rates, athough the BC study found a reduction for large firms and in the service sector. In Alberta, a study using aggregate data at the level of construction sub-industries found that inspections had no effect on the risk of lost-time injuries but did have a positive effect on reducing the number of work-related fatalities.[6]

In addition to the lack of empirical studies on the effectiveness of regulatory inspections in Canada, with some exceptions,[4,7–9] the majority of existing studies have focused on the effectiveness of regulatory inspections for firms in the manufacturing sector.[5,10–12] Although this can be justified because manufacturing firms have more stable (and better-defined) locations therefore providing better precision that injuries occurred on that site as opposed to elsewhere,[11] there are important sectoral differences in OHS prevention initiatives and regulatory enforcement. For example, the degree to which a worksite is stable or fixed varies by industry and therefore both OHS initiatives and the effectiveness of regulatory inspections may differ by sector. This is evident in the previously mentioned BC study that only found effectiveness of inspections in some sectors,[4] as well as a study on OHSA inspections that found that the mean effect of an inspection on worker safety was more evident in the manufacturing sector, compared to health services and other industries (albeit with estimates that were not statically significant).[8]

Another common limitation of existing studies on the effectiveness of regulatory inspections is the focus on firms with more than 10 employees due to the previously mentioned OHSA not conducting programmed inspections at firms with less than 10 employees.[11] Including firms of different sizes and accounting for firm size is important in terms of both OHS performance and the effectiveness of regulatory inspections. Compared to larger firms, smaller firms might be expected to receive lower savings from preventing injuries as they are subject to little or no experience rating by workers' compensation insurers, employ workers with a higher risk tolerance who may not put as much pressure on employers to improve safety, less able to realize economies of scale in production of safety, and less likely to be inspected, reducing the marginal benefit of compliance.[13] In contrast, research on the effect of OHSA inspections with penalties found that there was a weaker preventive effect of inspections in the very largest firms (500+ employees) than in smaller establishments.[14]

In response to the limited body of research on the effectiveness of regulatory inspections on firm-level injury rates in Canada and the lack of attention elsewhere to the effectiveness of inspections in sectors other than manufacturing or for firms of different sizes, the objective of this study was to use a sector-specific approach in a Canadian jurisdiction to address the following questions:

(1) Do firm-level injury rates decrease after regulatory inspections?
(2) Do specific regulatory inspections, such as multiple inspections or inspections with orders, lead to greater reductions in firm-level injury rates?

## 2 | METHODS

### 2.1 | OHS regulation and enforcement in Alberta

In Alberta, the organization responsible for workplace safety regulation is the Government of Alberta (GoA) Ministry of Jobs, Economy and Trade, and the organization responsible for workers' compensation is the Workers Compensation Board of Alberta (WCB). OHS officers have the authority to enter provincially regulated work sites, conduct inspections, and require these work sites to take steps to comply with the requirements of the OHS Act, Regulation and Code.[15] Each year, OHS officers proactively inspect firms to monitor compliance with legislation. Firms and sectors are chosen for inspections based on having relatively high injury and illness rates, high frequency of OHS incidents or complaints, and persistently low rates of compliance or emerging trends for these preceding criteria.

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In contrast to inspections, OHS officers may also conduct investigations that are more reactive and in response to workplace fatalities or major incidents.

During the past decade, OHS legislation and regulation in Alberta have undergone several changes to increase the power of OHS officers and introduce greater financial penalties for noncompliance. In 2012, the OHS Act was amended to allow OHS officers to request identifying information of workers and employers at work sites and service orders electronically.[16] As of October 1, 2013, the OHS department has had the ability to impose administrative penalties of up to $10,000 per day against any parties (employer, worker, prime contractor, supplier) for contraventions of the OHS legislation. As of January 1, 2014, OHS officers have had the ability to issue tickets (on-the-spot penalties of up to a maximum of $500) to workers and employers in contravention of ticketable provisions of OHS legislation.[17] Additional changes were introduced in June 1, 2018, however these fall outside the scope of this study.[18]

## 2.2 | Data

This study used data from the WCB and GoA for the years 2011–2018. WCB claim and firm-level data were linked using the WCB account number and industry operation identifier (ID) and used to create a unique firm ID. GoA regulatory enforcement data were extracted from the Compliance Management Information System (CMIS) database, in which client contact activities could be linked to the same WCB account numbers and industry operation IDs. Client contact activities included inspections, investigations, and meetings, and these could also include one or more deterrence measures such as orders, tickets and administrative penalties. Since date variables were available for both claim data (e.g., accident date) and regulatory inspections (e.g., inspection date), it was possible to construct cohorts of firms with details on the timing of injuries in relation to regulatory inspections (e.g., in the month prior or the year prior). While the CMIS database provided in-depth information on various regulatory activities, this study focused exclusively on the role of inspections and orders issued during them, as there were too few financial penalties or tickets issued to analyze them as a distinct enforcement activity.

## 2.3 | Cohort creation

To maximize the use of the data for the 8 years available, repeated cross-sections of firms were created in which firms could be in the treatment (inspected firm) or control (non-inspected firm) groups at different periods of time. Each cross-section covered a 37-month time frame divided into four distinct periods: a 12-month preintervention period, a 1-month intervention period (when the firm experienced a regulatory inspection or not), and two 12-month postintervention periods (see Figure 1 for illustration). This length of follow-up was chosen as research has shown how the impact of inspections varies over follow-up time[7] and is similar to that used in other Canadian research.[5]

For each period, firm and claim-level compensation data and regulatory enforcement activity data were extracted from WCB and MoL databases for each firm ID. To create comparable cohorts of treatment (inspected firms) and control (non-inspected firms) groups, and to minimize bias, the following inclusion criteria were applied. First, firms had to have at least one or more full-time equivalent employees (FTE) for 37 consecutive months. This ensured that there would be sufficient pre- and post-inspection periods to estimate the intervention effects. Second, firms had to have a preintervention baseline. In other words, inspected firms could not have had any inspections or other regulatory activities during the 12 months before the month of inspection. Third, firms could not have investigations taking place during the same month as the inspection because, unlike inspections, which are mainly proactive, investigations are reactive, incident/injury driven, and therefore show different patterns in injury as a result of them, compared to inspections.[11] A summary of cohort restrictions and firm-year-months is available in Supporting Information S1: Table A.

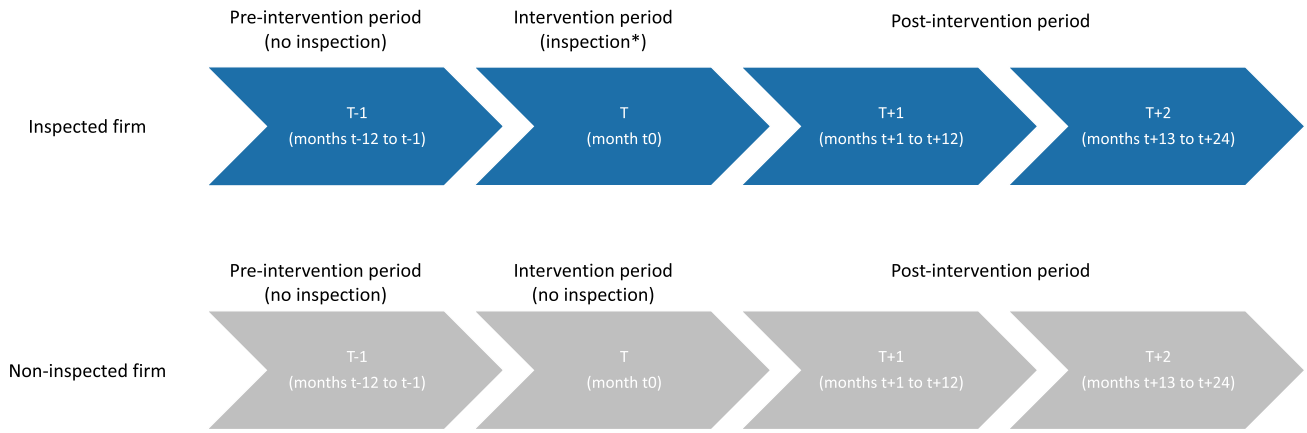


**FIGURE 1** Illustration of cross-sections of inspected and non-inspected firms by intervention period. *Inspections may include single inspections or multiple inspections, inspections without orders and inspections with orders (general and sector-specific).

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## 2.4 | Model specification

Changes in pre- and postintervention firm-level injury rates for treatment and control firms were compared using difference-in-differences (DiD) negative binomial regression models with the log of firm FTEs included as an offset variable and clustered by the firm to account for correlation of observations within firms over time. DiD analysis has been used in previous research on the effectiveness of inspections,[7] as has the use of negative binomial regression with a log of FTEs.[5] Accordingly, a baseline indicator measured the preintervention differences in the injury rate between firms that were inspected in month 0 versus those that were not. Two postintervention period indicators compared the injury rate of all firms 12 months after the intervention period and 12 months after that. The intervention effect was estimated by interacting the baseline indicator and the postintervention period indicator which provides an estimate of the relative change in the injury rate on inspected firms compared to the non-inspected firms.

Firm characteristics were balanced between the treatment group and control groups at the time of the intervention using Coarsened Exact Matching (CEM).[19] A one-to-many match was conducted for each inspected firm-year-month to as many eligible non-inspected firm-year months with the same observable characteristics during month 0. Inspected firm-year months were matched to non-inspected firm-year months based on observable firm-level characteristics identified in multivariable logit prediction models and model fit statistics of the Akaike information criterion and Bayesian information criterion. The following variables were used for matching: insurable earnings (categorical, quartiles); firm size (categorical); industry premium rate; experience rate adjustment (categorical: discounted, unadjusted, surcharged); industry subsector (derived from SIC 1980 division, major and minor group codes); rate year; and finally, a binary indicator of whether the firm had a lost-time injury claim in the previous 12 months. All but the final matching variable were included in the final regression models, in addition to firm tenure (categorical); whether the firm was a member of the Partnerships in Injury Reduction (PIR), a voluntary OHS program provided by the WCB and GoA; whether they had a Certificate of Recognition (COR), a voluntary audit-based OHS management system certification[20]; and whether their head office location was in Alberta. Matching variables were included in the regression models due to their time-varying nature in that all of them except the industry subsector could take different values during the follow-up period compared to the baseline.

A sector-specific approach was taken for two main reasons: first, injury prevention measures and regulatory enforcement activities are typically industry-based (e.g., fall protection orders are specific to the construction sector). Second, due to changes in the way WCB firm accounts were classified in the data (particularly in the healthcare sector), it was not possible to conduct a study of the overall effectiveness of inspections across all industries in the province. The construction, manufacturing, and transportation sectors were chosen since they all experienced injury claim rates above the provincial average[21] as well together accounting for 70% of all inspections in the CMIS data. For each sector, the overall effect of having an inspection was estimated, as well as the effect of having more than one inspection, an inspection with any order, an inspection with stop use/work order, and an inspection with an industry-specific order based on the most common legislation order within the sector (Supporting Information S1: Table B). To obtain the intervention effect for these specific types of inspections, the linear combination of estimates from the separate interaction terms was calculated. The effect estimates do not differentiate between whether tickets and administrative penalties were also issued with the inspections with orders.

All claims receiving compensation for work-related injuries and musculoskeletal diseases were included in the analyses. Analysis was conducted for lost-time claims, which are accepted claims for work-related injury or musculoskeletal disease, with work disability (compensated lost-time) beyond the day of injury. Occupational cancers and mental health condition claims were excluded due to their long latency periods and to focus on the types of injuries that would more likely be affected by inspections. Sensitivity analyses were also conducted for disabling injury claims (accepted claims for work-related injury or musculoskeletal disease with either lost-time from work beyond the day of injury or return to work but with modified work duties) as well as for small and large firms using a threshold of <20 FTEs and 20+ FTEs to differentiate small from large firms based on legislative requirements.[22] Data preparation and analyses were conducted using Stata 16.1 (StataCorp.). Access to data was provided under data access agreements between the University of British Columbia, WCB, and MoL.

# 3 | RESULTS

Summary statistics of the unmatched inspected and non-inspected firms during the months they were eligible for inspection are shown in Table 1. Overall, the number of firm-year-months with inspections was highest in the construction sector (3835), followed by the manufacturing and transportation sectors (1154 and 303, respectively). Across all three sectors, compared to non-inspected firm-year-months, inspected firm-year-months were associated with larger firms, older firms, higher insurable earnings, insurance premium surcharges, fewer PIR members, and fewer COR certifications. Following CEM matching, firm-year-month characteristics were more balanced (Supporting Information S1: Table C).

Using the fully matched cohorts, the lost-time claim rate of inspected firms was 8% higher than non-inspected firms at baseline in the construction sector (IRR: 1.08, 95% CI: 1.01–1.14) and manufacturing sectors (IRR: 1.08, 95% CI: 0.99–1.17) but no different in the transportation sector (IRR: 1.01, 95% CI: 0.84–1.20) (Table 2). In the construction sector, inspected firms were associated with a 12% increase in the lost-time claim rate (IRR: 1.12, 95% CI: 1.04–1.21) relative to the change in non-inspected firms during the first year, but no different in the second year (IRR: 1.02, 95% CI: 0.94–1.10), postintervention. In contrast, in the postintervention period, the injury rate decreased overall in construction in the first (IRR: 0.92 [0.90–0.94]) and second (IRR: [0.96-0.98]) years, meaning that inspected firms had a higher baseline rate that increased slightly, while non-inspected firms experienced a decline. In the manufacturing and transportation sectors, no inspection intervention effect was observed for either the first or second years postintervention.

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**TABLE 1** Summary statistics for inspected and non-inspected firm-year-months in the construction, manufacturing, and transportation sectors before matching.

| | Construction | | Manufacturing | | Transportation | |
|---|---|---|---|---|---|---|
| | Inspected (3835) | Non-inspected (883,535) | Inspected (1154) | Non-inspected (313,796) | Inspected (303) | Non-inspected (263,509) |
| Insurable earnings quartiles | | | | | | |
| 1st | 6.4 | 25.1 | 1.1 | 25.1 | 3.3 | 25.0 |
| 2nd | 15.3 | 25.0 | 5.8 | 25.1 | 7.3 | 25.0 |
| 3rd | 23.8 | 25.0 | 22.2 | 25.0 | 19.8 | 25.0 |
| 4th | 54.5 | 24.9 | 70.9 | 24.8 | 69.6 | 24.9 |
| Firm size (FTEs) | | | | | | |
| 1–4 | 28.5 | 60.0 | 5.7 | 47.0 | 12.5 | 54.0 |
| 5–9 | 18.4 | 16.6 | 10.9 | 17.4 | 12.2 | 18.4 |
| 10–19 | 16.5 | 10.8 | 18.6 | 13.9 | 18.5 | 11.8 |
| 20–99 | 26.9 | 10.3 | 43.8 | 17.3 | 41.3 | 13.5 |
| 100+ | 9.7 | 2.2 | 21.0 | 4.4 | 15.5 | 3.3 |
| Industry subsector | | | | | | |
| Sash/door and other millwork | 1.9 | 2.0 | | | | |
| Residential building and development | 28.5 | 13.1 | | | | |
| Nonresidential building and development | 8.7 | 5.3 | | | | |
| Industrial construction | 2.2 | 3.7 | | | | |
| Highway and heavy construction | 12.2 | 10.6 | | | | |
| Site work | 2.6 | 6.9 | | | | |
| Structural and related work | 5.0 | 4.3 | | | | |
| Exterior close-in work | 15.5 | 3.4 | | | | |
| Plumbing/heating and air conditioning/ mechanical work | 4.8 | 7.6 | | | | |
| Mechanical specialty work | 1.9 | 2.9 | | | | |
| Electrical work | 3.1 | 8.5 | | | | |
| Interior and finishing work | 10.0 | 12.9 | | | | |
| Other trade work | 1.1 | 1.8 | | | | |
| Services incidental to construction | 0.4 | 9.5 | | | | |
| Electrical machinery/equipment and supplies/wholesale | 0.3 | 2.4 | | | | |
| Architectural/engineering and other scientific and technical services | 1.1 | 3.5 | | | | |
| Miscellaneous industries | 0.7 | 1.6 | | | | |
| Food and beverage industries | | | 7.3 | 7.6 | | |
| Rubber products industries | | | 4.4 | 2.7 | | |
| Textile products/clothing industries | | | 1.4 | 5.1 | | |
| Wood/furniture industries | | | 13.9 | 9.5 | | |
| Paper/printing/publishing and allied industries | | | 2.2 | 5.2 | | |
| Primary/fabricated metal products industries | | | 46.4 | 27.2 | | |

(Continues)

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**TABLE 1** (Continued)

| | Construction | | Manufacturing | | Transportation | |
|---|---|---|---|---|---|---|
| | Inspected (3835) | Non-inspected (883,535) | Inspected (1154) | Non-inspected (313,796) | Inspected (303) | Non-inspected (263,509) |
| Machinery/electrical industries | | | 2.3 | 4.7 | | |
| Non-metallic mineral products | | | 7.9 | 2.7 | | |
| Refined petro and coal/chemical/and other manufacturing | | | 4.9 | 6.3 | | |
| Building/developing and general/trade contracting | | | 5.6 | 23.2 | | |
| Miscellaneous industries | | | 3.9 | 5.7 | | |
| | Construction | | Manufacturing | | Transportation | |
| | Inspected (3835) | Non-inspected (883,535) | Inspected (1133) | Non-inspected (82,657) | Inspected (295) | Non-inspected (81,828) |
| Non-truck transportation | | | | | 5.6 | 10.8 |
| Truck transportation | | | | | 47.2 | 72.9 |
| Storage and warehousing/other services incidental to transportation | | | | | 8.3 | 1.6 |
| Other utility industries | | | | | 12.9 | 4.7 |
| Waste materials/wholesale | | | | | 14.9 | 3.0 |
| Other motor vehicle services | | | | | 6.6 | 1.5 |
| Miscellaneous industries | | | | | 4.6 | 5.6 |
| Industry premium rate | 3.3 | 2.3 | 1.9 | 1.7 | 3.0 | 3.3 |
| Experience rate adjustment | | | | | | |
| Discounted | 52.5 | 56.7 | 48.9 | 59.1 | 51.2 | 61.8 |
| Unadjusted | 18.0 | 30.4 | 6.8 | 24.4 | 11.9 | 23.5 |
| Surcharged | 29.5 | 13.0 | 44.3 | 16.5 | 37.0 | 14.7 |
| Rate year | | | | | | |
| 2012 | 19.8 | 19.1 | 22.4 | 19.7 | 19.5 | 20.1 |
| 2013 | 20.8 | 20.2 | 19.2 | 20.5 | 17.2 | 20.3 |
| 2014 | 19.0 | 20.2 | 19.2 | 20.3 | 15.8 | 20.1 |
| 2015 | 19.9 | 20.4 | 21.7 | 19.8 | 24.1 | 19.9 |
| 2016 | 20.5 | 20.0 | 17.6 | 19.6 | 23.4 | 19.6 |
| Firm tenure (years) | | | | | | |
| 0–5 | 28.9 | 33.2 | 15.8 | 25.2 | 21.5 | 26.5 |
| 6–10 | 23.7 | 24.3 | 17.1 | 22.3 | 22.8 | 23.1 |
| 11–20 | 23.9 | 24.7 | 27.3 | 25.7 | 28.7 | 27.1 |
| 21+ | 23.5 | 17.7 | 39.9 | 26.9 | 27.1 | 23.3 |
| PIR member | | | | | | |
| Yes | 8.5 | 2.7 | 15.0 | 3.7 | 4.6 | 1.8 |
| No | 91.5 | 97.3 | 85.0 | 96.3 | 95.4 | 98.2 |
| COR-certified | | | | | | |
| Yes | 38.4 | 21.5 | 39.3 | 14.2 | 43.2 | 21.0 |
| No | 61.6 | 78.5 | 60.7 | 85.8 | 56.8 | 79.0 |

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**TABLE 1** (Continued)

| | Construction | | Manufacturing | | Transportation | |
|---|---|---|---|---|---|---|
| | Inspected (3835) | Non-inspected (883,535) | Inspected (1133) | Non-inspected (82,657) | Inspected (295) | Non-inspected (81,828) |
| Employer address in Alberta | | | | | | |
| Yes | 94.1 | 92.7 | 87.5 | 91.7 | 81.8 | 87.4 |
| No | 5.9 | 7.3 | 12.5 | 8.3 | 18.2 | 12.6 |

*Note*: Column percentages for the inspected firms are based on the month in which they were inspected, while for the non-inspected firms, the percentages reflect the share of firm-year-months for months in which they were eligible for inspection.

Abbreviations: COR, certificate of recognition; FTE, full-time equivalent employee; PIR member, partnerships in injury reduction member.

**TABLE 2** Claim rate ratios with 95% confidence intervals of the effect of inspections on firm-level lost-time injury rates using matched cohorts by sector.

| | Construction | | Manufacturing | | Transportation | |
|---|---|---|---|---|---|---|
| | RR | 95% CI | RR | 95% CI | RR | 95% CI |
| Baseline difference | 1.08 | (1.01–1.14) | 1.08 | (0.99–1.17) | 1.01 | (0.84–1.20) |
| Postintervention period (t + 1) | 0.92 | (0.90–0.94) | 0.95 | (0.92–0.98) | 1.05 | (0.98–1.12) |
| Postintervention period (t + 2) | 0.96 | (0.93–0.98) | 0.89 | (0.86–0.93) | 1.12 | (1.04–1.20) |
| Intervention effect (t + 1) | 1.12 | (1.04–1.21) | 1.01 | (0.91–1.11) | 1.07 | (0.88–1.31) |
| Intervention effect (t + 2) | 1.02 | (0.94–1.10) | 1.00 | (0.90–1.11) | 0.92 | (0.75–1.13) |
| Firm-years | 1,369,704 | | 251,370 | | 246,369 | |

*Note*: Firms were matched during the intervention month on insurable earnings, firm size, industry premium rate, experience rate adjustment, and industry subsector using a one-to-many coarsened exact matching. Postintervention periods represent the postintervention periods for both firms that were inspected during month t0 and firms that were not inspected during month t0. Firm years represent 12-month periods prior (t−1) to the intervention month (t0), the 12-month period after that (t + 1), and the 12-month period after that (t + 2). Models were adjusted for insurable earnings, firm size, industry premium rate, experience rate adjustment, industry subsector, rate year, firm tenure, firm membership in Partnerships in Injury Reduction, Certificate of Recognition, and employer address location.

Table 3 shows the effect estimates of specific types of inspections on changes in the lost-time claim rate. In the construction sector, having more than one inspection was not associated with a decrease in the lost-time claim rate in the first or second year, postintervention, compared to non-inspected firms during the same period. Inspections with orders were associated with a 16% increase in the lost-time claim rate during the first year but no difference in the second year, postintervention. Inspections with stop work/use orders were not associated with any differences in the changes in the lost-time claim rate. Inspections with fall protection orders were associated with a 22% increase in the injury rate during the first year but no difference in the second year relative to the change in non-inspected firms postintervention.

For the transportation sector, a reduction in the injury rate due to various inspection activities was found in the second year, postintervention. However, the only precisely estimated difference was observed for firms with more than one inspection which was associated with a 28% reduction in the lost-time claim rate. No intervention effect was found for the manufacturing sector estimates, regardless of the type of inspection, with point estimates all close or equal to one, with confidence intervals crossing one.

In the sensitivity analyses, the overall pattern of findings for disabling injury claim rates was similar (Supporting Information S1: Table D), as were the findings for small and large firms (Supporting Information S1: Tables E and F). However, there was evidence of increased lost-time claim rates among inspected small construction firms relative to non-inspected small construction firms during the same period. For example, inspections among this group were associated with a 25% increase in the lost-time claim rate during the first year and a 15% increase during the second year, postintervention.

## 4 | DISCUSSION

### 4.1 | Key findings

In general, regulatory workplace safety inspections were not associated with greater reductions in lost-time claim rates among

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**TABLE 3** Claim rate ratios with 95% confidence intervals of the effect of specific inspection activities on firm-level lost-time injury rates using matched cohorts by sector.

| | Model I | | Model II | | Model III | | Model IV | | Model V | |
|---|---|---|---|---|---|---|---|---|---|---|
| | RR | 95% CI | RR | 95% CI | RR | 95% CI | RR | 95% CI | RR | 95% CI |
| Construction | | | | | | | | | | |
| Inspected (t0) | 1.08 | (1.01–1.14) | 0.91 | (0.83–0.99) | 1.03 | (0.94–1.12) | 1.03 | (0.94–1.12) | 1.03 | (0.94–1.12) |
| Inspected more than once (t0) | | | 1.43 | (1.27–1.63) | | | | | | |
| Inspected with orders issued (t0) | | | | | 1.10 | (0.97–1.23) | 1.08 | (0.94–1.23) | 1.11 | (0.97–1.27) |
| Inspected with stop work/use orders issued (t0) | | | | | | | 1.05 | (0.89–1.23) | | |
| Inspected with fall protection orders issued (t0) | | | | | | | | | 0.96 | (0.82–1.14) |
| Postintervention period (t + 1) | 0.92 | (0.90–0.94) | 0.93 | (0.91–0.94) | 0.92 | (0.90–0.94) | 0.92 | (0.90–0.94) | 0.92 | (0.90–0.94) |
| Postintervention period (t + 2) | 0.96 | (0.93–0.98) | 0.96 | (0.94–0.98) | 0.96 | (0.93–0.98) | 0.96 | (0.93–0.98) | 0.96 | (0.93–0.98) |
| Intervention effect (t + 1)[a] | 1.12 | (1.04–1.21) | 1.08 | (0.98–1.18) | 1.16 | (1.05–1.28) | 1.12 | (0.95–1.31) | 1.22 | (1.03–1.44) |
| Intervention effect (t + 2)[a] | 1.02 | (0.94–1.10) | 0.94 | (0.84–1.04) | 1.06 | (0.95–1.18) | 1.02 | (0.85–1.22) | 1.03 | (0.86–1.24) |
| Firm-years | 1,369,704 | | | | | | | | | |
| Manufacturing | | | | | | | | | | |
| Inspected (t0) | 1.08 | (0.99–1.17) | 0.95 | (0.84–1.07) | 1.01 | (0.88–1.15) | 1.01 | (0.88–1.15) | 1.01 | (0.88–1.15) |
| Inspected more than once (t0) | | | 1.25 | (1.08–1.45) | | | | | | |
| Inspected with orders issued (t0) | | | | | 1.11 | (0.95–1.31) | 1.10 | (0.93–1.30) | 1.08 | (0.91–1.29) |
| Inspected with stop work/use orders issued (t0) | | | | | | | 1.13 | (0.84–1.53) | | |
| Inspected with PPE orders issued (t0) | | | | | | | | | 1.08 | (0.88–1.33) |
| Postintervention period (t + 1) | 0.95 | (0.92–0.98) | 0.94 | (0.92–0.97) | 0.95 | (0.92–0.98) | 0.95 | (0.92–0.98) | 0.95 | (0.92–0.98) |
| Postintervention period (t + 2) | 0.89 | (0.86–0.93) | 0.89 | (0.85–0.92) | 0.89 | (0.86–0.93) | 0.89 | (0.86–0.93) | 0.89 | (0.86–0.93) |
| Intervention effect (t + 1)[a] | 1.01 | (0.91–1.11) | 1.03 | (0.91–1.16) | 0.99 | (0.87–1.12) | 1.06 | (0.74–1.50) | 1.08 | (0.87–1.33) |
| Intervention effect (t + 2)[a] | 1.00 | (0.90–1.11) | 1.04 | (0.91–1.18) | 0.98 | (0.86–1.12) | 1.07 | (0.77–1.47) | 1.03 | (0.81–1.30) |
| Firm-years | 251,370 | | | | | | | | | |
| Transportation | | | | | | | | | | |
| Inspected (t0) | 1.01 | (0.84–1.20) | 0.82 | (0.65–1.02) | 0.94 | (0.71–1.24) | 0.94 | (0.71–1.24) | 0.94 | (0.71–1.24) |
| Inspected more than once (t0) | | | 1.62 | (1.18–2.23) | | | | | | |
| Inspected with orders issued (t0) | | | | | 1.12 | (0.80–1.58) | 1.13 | (0.80–1.59) | 1.20 | (0.83–1.73) |
| Inspected with stop work/use orders issued (t0) | | | | | | | 0.92 | (0.43–1.97) | | |
| Inspected with hazard orders issued (t0) | | | | | | | | | 0.86 | (0.56–1.33) |
| Postintervention period (t + 1) | 1.05 | (0.98–1.12) | 1.06 | (1.00–1.13) | 1.05 | (0.98–1.12) | 1.05 | (0.98–1.12) | 1.05 | (0.98–1.12) |
| Postintervention period (t + 2) | 1.12 | (1.04–1.20) | 1.14 | (1.06–1.23) | 1.12 | (1.04–1.20) | 1.12 | (1.04–1.20) | 1.12 | (1.04–1.20) |
| Intervention effect (t + 1)[a] | 1.07 | (0.88–1.31) | 0.90 | (0.68–1.19) | 1.15 | (0.92–1.44) | 1.40 | (0.60–3.24) | 1.34 | (0.92–1.96) |
| Intervention effect (t + 2)[a] | 0.92 | (0.75–1.13) | 0.72 | (0.53–0.99) | 0.87 | (0.67–1.12) | 1.43 | (0.41–4.94) | 1.00 | (0.69–1.44) |
| Firm-years | 246,369 | | | | | | | | | |

*Note*: Firms were matched during the intervention month on insurable earnings, firm size, industry premium rate, experience rate adjustment, and industry subsector using a one-to-many coarsened exact matching. Postintervention periods represent the postintervention periods for both firms that were inspected during month t0 and firms that were not inspected during month t0. Firm-years represent 12-month periods prior (t−1) to the intervention month (t0), the 12-month period after that (t + 1), and the 12-month period after that (t + 2). Models were adjusted for insurable earnings, firm size, industry premium rate, experience rate adjustment, industry subsector, rate year, firm tenure, firm membership in Partnerships in Injury Reduction, Certificate of Recognition, and employer address location.

[a]The main intervention effects for inspections (Model I) were estimated using an interaction term between the baseline (inspected (t0) and postintervention periods (t + 1 or t + 2). The intervention effects for specific types of inspections (Models II–V) were estimated using the linear combination of estimates of the separate interaction terms for each inspection type.

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inspected firms compared to similar non-inspected firms in the Alberta construction, manufacturing, and transportation sectors. Across the 15 statistical models, the only precisely estimated example of a greater injury rate reduction due to inspection was for transportation firms having more than one inspection and experiencing a greater reduction during the second year, post-inspection. In the construction sector, inspections, inspections with orders, and inspections with fall protection orders were associated with a greater increase in the lost-time claim rate compared to non-inspected firms during the first year, post-inspection. Furthermore, this increase in lost-time claim rate was higher when focusing on small construction firms. No intervention effect for inspections was observed in the manufacturing sector.

## 4.2 | Comparisons to previous research

The main study findings show consistency with previous research in Canada despite the use of different study designs and statistical methods. Research in Ontario found that there were no significant differences in claim rates between manufacturing firms targeted for consultations, inspections, or no intervention during the 2 years following the interventions.[5] Research in Alberta using construction subindustry data found that inspections had no effect on the risk of lost-time injuries but did have a positive effect in reducing the number of work-related fatalities.[6] Research in BC found that inspections overall did not result in reductions in the claim rate, but inspections among service sector firms did result in reductions in injuries during the first 12 months following inspection.[4] In contrast, several studies in the United States have found evidence of an overall effectiveness of inspections, particularly those with penalties, in reducing firm-level injury rates, whether it was in Washington State,[9] California,[7] or multiple states in which OSHA inspections have been conducted.[14] These divergent trends may be explained by the fact that despite inspections being less common in US jurisdictions, they are more likely to be accompanied by punitive measures, such as penalties, and therefore lead to greater deterrence.[4] Further, as inspections are less common in US jurisdictions compared to Canada, they may focus on more recalcitrant firms than in Canada such that improvements in the injury rate are more readily achieved.[4]

Previous research in BC found that inspections with orders were only effective in reducing claim rates among service sector firms during the first year following inspection.[4] Research in Washington State found that inspections with citations were associated with a 20% decline in non-MSD claim rates relative to uninspected workplaces, whereas inspections without citations were associated with a smaller nonsignificant decline.[9] Had this study been able to separate MSD from non-MSD lost-time, it is possible that inspections may have shown more effectiveness. Research on OHSA inspections across various states,[12] and a study specific to Pennsylvania,[11] found that there were reductions in claim rates for inspections citing violations of the general requirement for PPE for manufacturing firms. There was no evidence in this study that inspections with orders, inspections with stop work/stop use, or industry-specific orders were more effective than inspections without orders. Nonetheless, it should be noted that the OHSA studies were restricted to inspections with penalties.

In some cases, such as in the construction sector, inspected firms experienced a greater increase in their claim rate relative to the change observed for non-inspected firms. While this supports the main finding that regulatory inspections are not effective in reducing firm-level injury rates, the reasons for an increased claim rate are unclear. A study on the effectiveness of OSHA inspections in the manufacturing sector found that non-penalty inspections were, in some cases, followed by increases in claims.[14] The authors of that study proposed that a clean bill of health from inspectors may reduce managerial attention to safety and prevention measures and thereby result in an increase in claims following inspection. Nonetheless, the current study observed an increased claim rate among inspected construction firms, even for firms with inspections with orders. As such, regulatory and compliance deficiencies were identified at these worksites. When the construction sector-specific models were stratified by firm size, the increased rate of firm-level injuries for inspected firms was driven by smaller firms. This suggests that whatever is leading to an increase in injury rates is more common among small firms. However, the theoretical and causal models underpinning this study are based on the hypothesis that inspections reduce injuries compared to not reducing injuries, and the observed increase in injury rates primarily supports the finding that the various inspection activities were not effective.

Findings from previous studies show substantial variation in the effectiveness of inspections after comparing inspections with and without penalties or citations. For example, among OSHA inspections across the United States, inspections with penalties have been more effective in reducing claims than inspections without penalties.[14] In the manufacturing sector in Pennsylvania, inspections with penalties reduced claims by an average of 19%–24% annually in the 2 years following inspection.[10] Administrative penalties and tickets have been part of OHS legislation and regulation in Alberta since 2013 and 2014, but there were too few of them to analyze in a meaningful way in this study. Given a longer study period and larger sample size, it may be that firms that experience inspections with administrative penalties or tickets result in greater injury rate reductions than similar non-inspected firms.

## 4.3 | Strengths and limitations

A study strength has been having access to and linkage of firm and claim-level workers' compensation data and regulatory enforcement activity data. This is the first study to link WCB data with the GoA data to examine whether firms that are inspected experience different firm-level injury outcomes compared to non-inspected firms with similar characteristics. Complementing the data, the

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observational research design using DiD and CEM matching offered a robust methodological approach to evaluate the effectiveness of inspections in the absence of a randomized research design, such as that used in studies in Ontario[5] and California.[7] DiD enabled the comparison of changes over time in the non-inspected control firms with inspected firms to estimate the effects of inspections had they not taken place. The CEM matching enabled the inspected firms to be matched with comparable non-inspected firms on baseline characteristics. The negative binomial regression models provided effect estimates that accounted for the correlation of observations within firms over time and weighted the model estimates appropriately to account for the large number of small firms with zero claim counts.

There are some study limitations that must be acknowledged. The main limitation is the retrospective and observational study design. While appropriate methods were used to imitate a randomized study design (i.e., DiD and matching), it may not be as rigorous as a randomized study. Another study limitation was the inability to examine the effectiveness of all of the different regulatory enforcement activities that are conducted by OHS officers in Alberta, such as cooperative measures of meetings, presentations, and punitive measures of administrative penalties and OHS tickets. It would be useful to test whether consultations are effective in reducing injuries, as is the case in Washington,[9] or not effective, like in Ontario,[5] or know whether inspections with penalties outperform inspections without penalties, as is the case with studies in the United States.[10,14] Similarly, it would be useful to have examined whether changes in regulatory activities over time, such as the introduction of OHS tickets, impacted the overall effectiveness of inspections.

This study only looked at the effectiveness of inspections with regard to reductions in lost-time and disabling injury claim rates. These outcomes were of key interest to the stakeholders of this study (WCB and GoA). Larger sample sizes may have enabled the investigation of the effectiveness of inspections and inspections with particular citations in relation to different injury types.[11,12] Another study limitation was the inability to apply a fixed effects model. A longer time window to study inspections for each firm would enable firms to be their own controls in models to minimize biases, as has been done in studies elsewhere.[4]

Firm-level claim rates were calculated using FTEs based on wage data that are sensitive to issues such as how suitable the average industry wage estimates are from Statistics Canada, the timeliness of the estimates, and how the total insurable earnings cap may artificially produce fractional FTEs and therefore inflate firm-level injury rates.

The final limitation to note is the inability to distinguish firms with single or multiple operating locations within the administrative data. Ideally, this study would have applied the same cohort restrictions as previous studies by focusing on firms with single operating locations to isolate the effect of inspections.[4,9,11] Nonetheless, applying such a restriction to the construction and transportation sectors may not make sense considering the operating location does not necessarily represent the worksite in these sectors as it would for manufacturing. Research in Washington State was able to compare the effectiveness of inspections and consultations between fixed and non-fixed worksites and found that a significant decline in injury rates as a result of consultations was evident in non-fixed worksites.[9] The authors of the Washington State study suggest a construction firm requesting consultation may use the results to improve safety across all of its worksites.

## 5 | CONCLUSION

The findings suggest that inspections alone may not be sufficient to induce compliance or hazard management changes that lead to reductions in firm-level claim rates in the Alberta construction, manufacturing, or transportation sectors. Inspections may need to be supplemented with additional support such as ensuring firms have the knowledge and resources to make meaningful health and safety changes and that these are linked to clearly defined and sustained changes in hazard management.[23] Alternatively, the GoA may need to apply stricter compliance measures, such as administrative penalties and tickets. Increased claim rates following inspections among construction firms, particularly small firms, warrants further research as it is unclear whether this change as a result of inspections is positive, in that firms may be improving their reporting of injuries, or negative, that firms may be paying less attention to injury prevention as a result of the inspection.

### AUTHOR CONTRIBUTIONS

Christopher B. McLeod, Robert A. Macpherson, and Ailin He conceived the study. Ailin He conducted the analysis. Christopher B. McLeod and Robert A. Macpherson guided the analysis and drafted the manuscript. All authors reviewed the data analysis, contributed to the manuscript preparation, and approved the final manuscript.

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### CONFLICTS OF INTEREST STATEMENT

The authors declare that there are no conflicts of interest.

### DISCLOSURE BY AJIM EDITOR OF RECORD

John Meyer declares that he has no conflict of interest in the review and publication decision regarding this article.

### DATA AVAILABILITY STATEMENT

Data may be obtained from the Workers' Compensation Board of Alberta and the Government of Alberta and are not publicly available. Data in the study were made available under an information sharing agreement from the respective data stewards.

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## ETHICS APPROVAL AND INFORMED CONSENT

Ethical approval for the research project was obtained from the Behavioural Research Ethics Board of the University of British Columbia (H18-00702).

## DISCLAIMER

All inferences, opinions, and conclusions drawn in this manuscript are those of the authors and do not reflect the opinions of the Alberta Ministry of Labour and Immigration or the Workers' Compensation Board of Alberta.

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